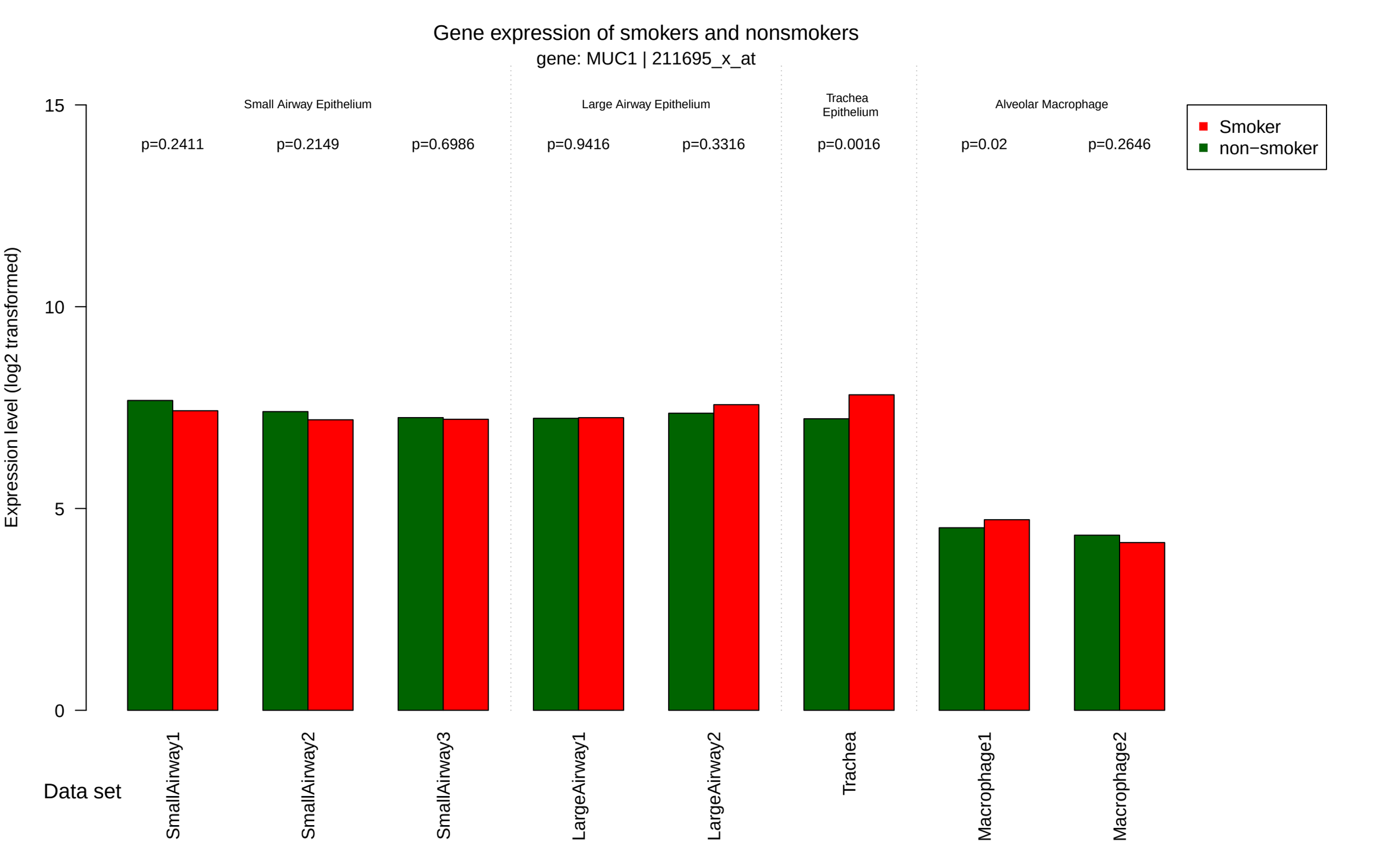
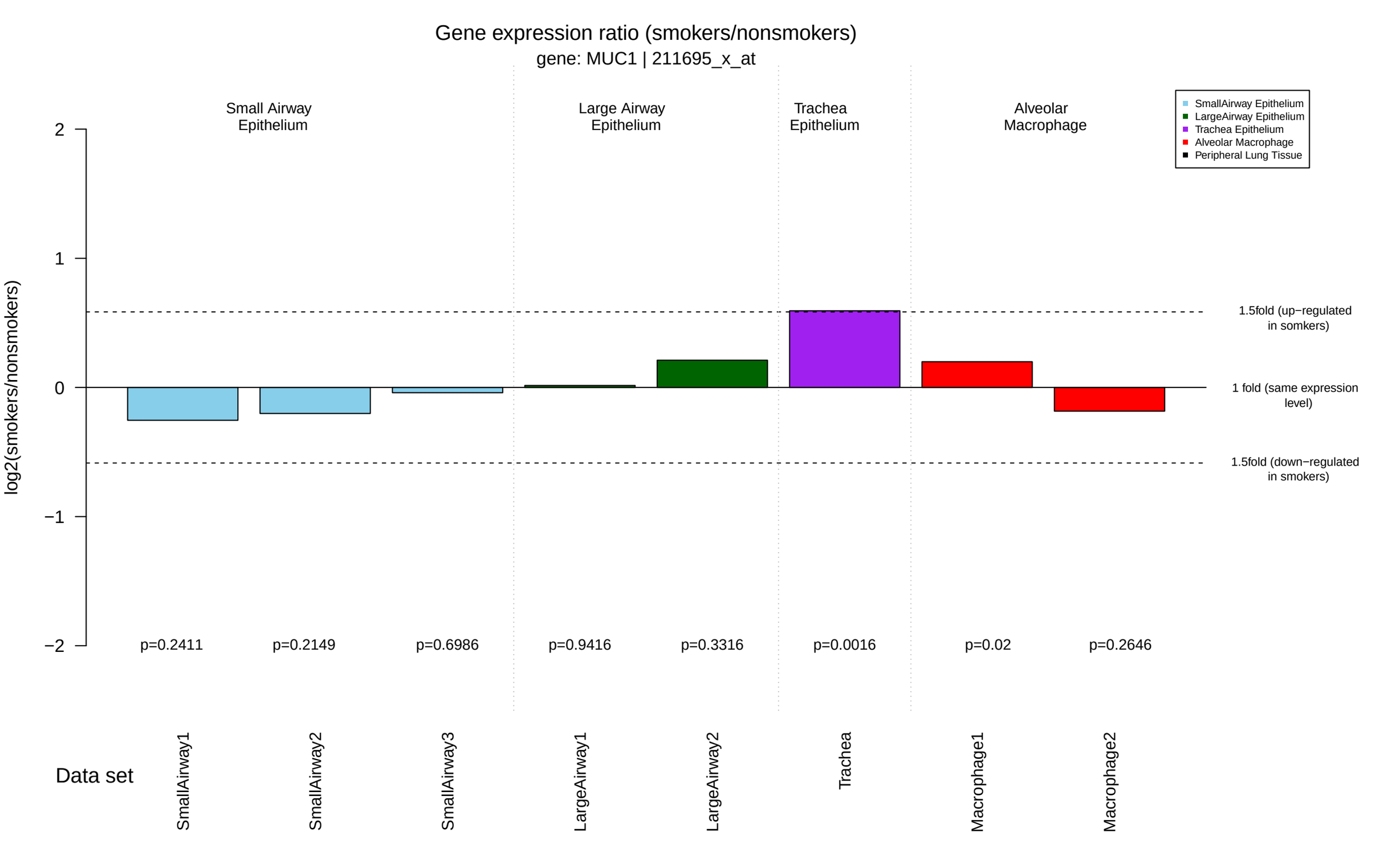
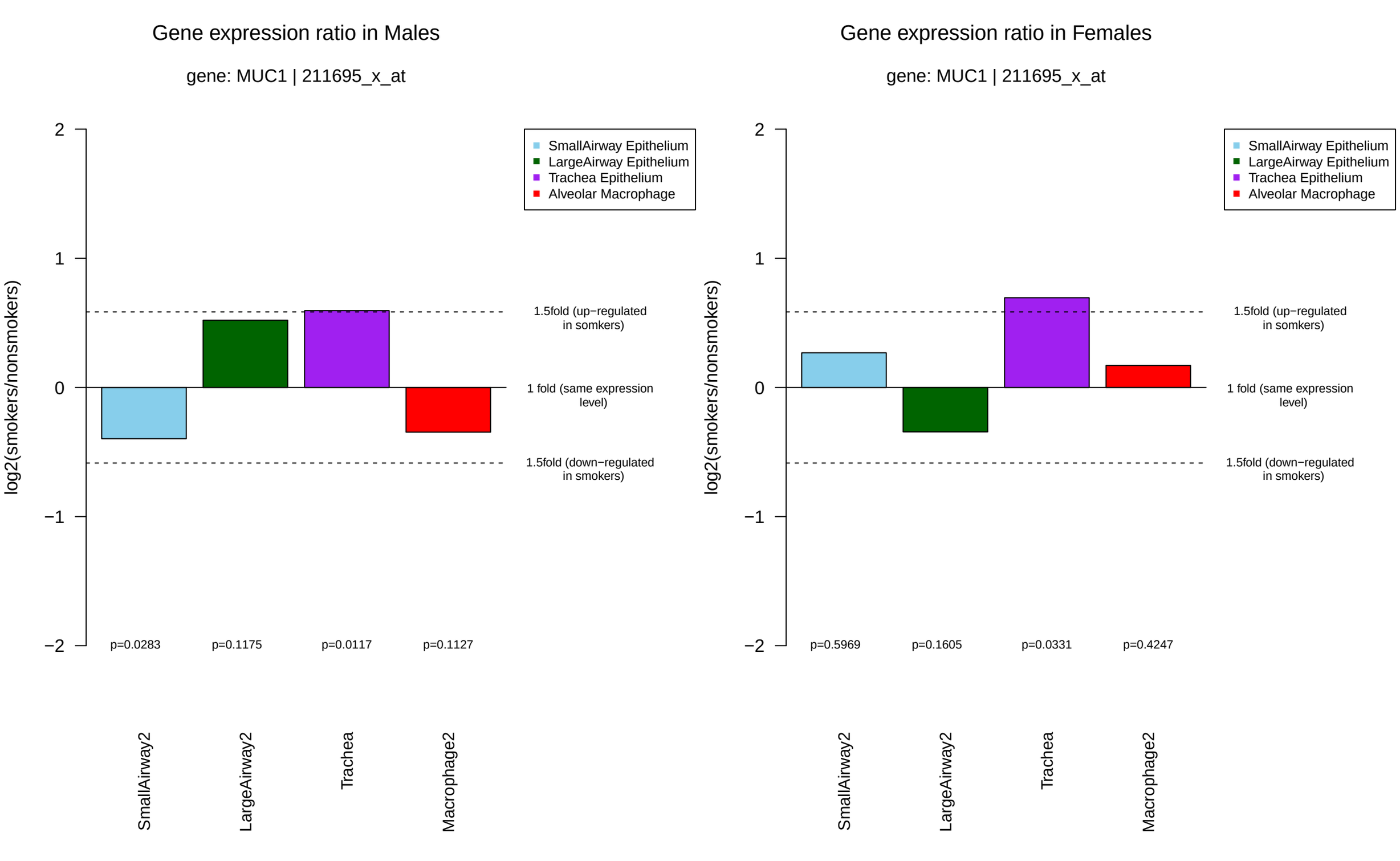
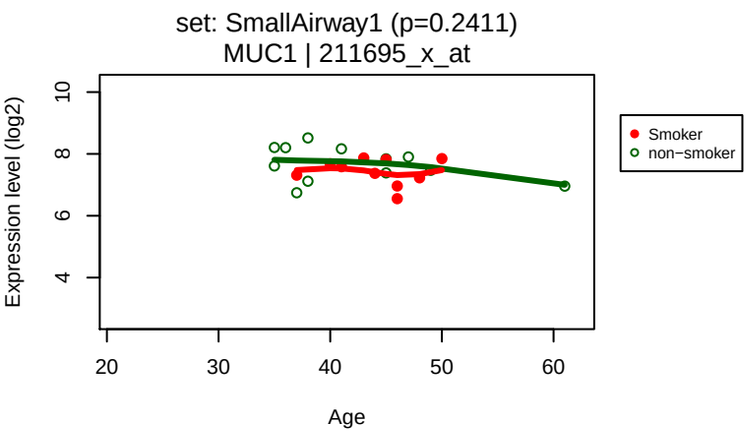
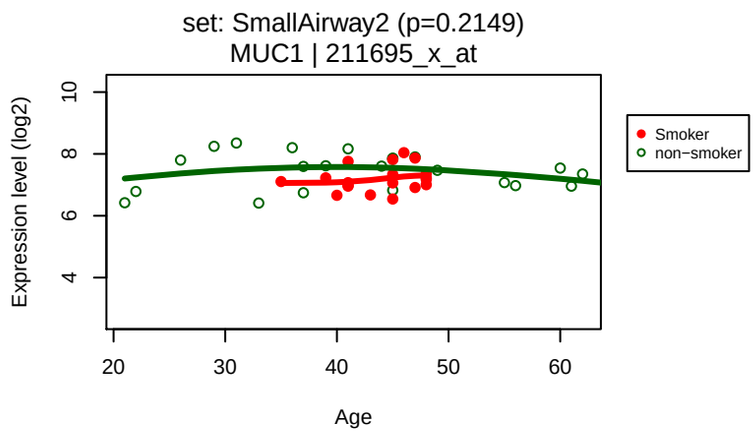
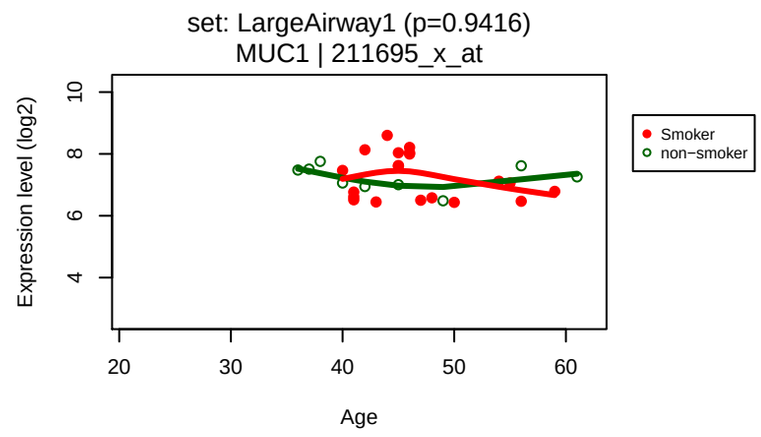

Gene expression of smokers and nonsmokers
gene: MUC1 | 211695_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2411
p=0.2149
p=0.6986
p=0.9416
p=0.3316
p=0.0016
p=0.02
p=0.2646
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

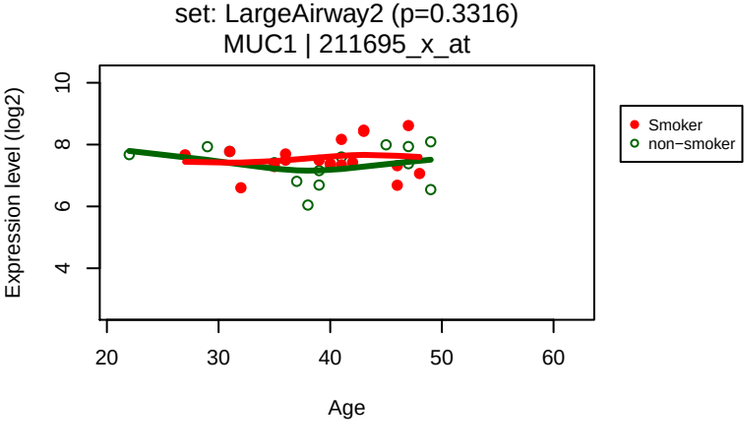

Gene expression ratio (smokers/nonsmokers)
gene: MUC1 | 211695_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2411
p=0.2149
p=0.6986
p=0.9416
p=0.3316
p=0.0016
p=0.02
p=0.2646
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

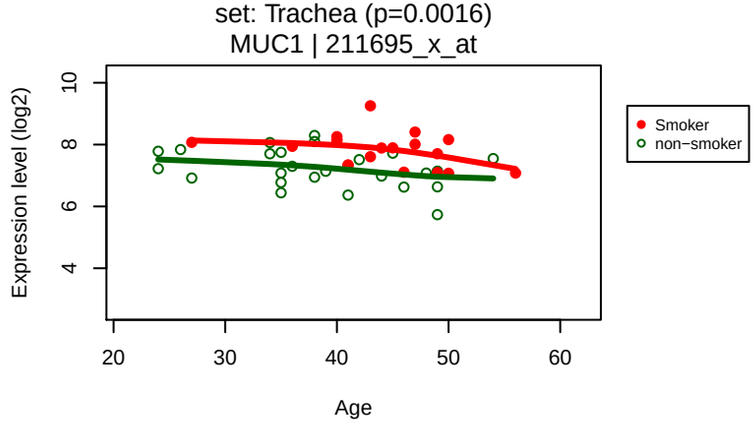

Gene expression ratio in Males
gene: MUC1 | 211695_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0283
p=0.1175
p=0.0117
p=0.1127
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC1 | 211695_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5969
p=0.1605
p=0.0331
p=0.4247
SmallAirway2
LargeAirway2
Trachea
Macrophage2

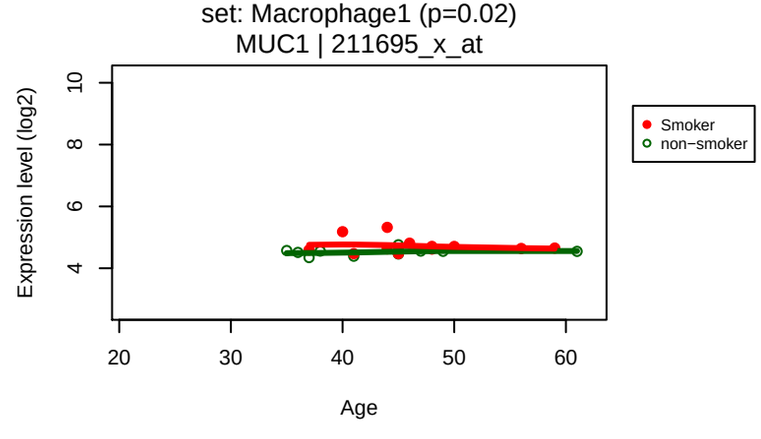

set: SmallAirway1 (p=0.2411)
MUC1 | 211695_x_at
Expression level (log2)
Age
Smoker
non−smoker

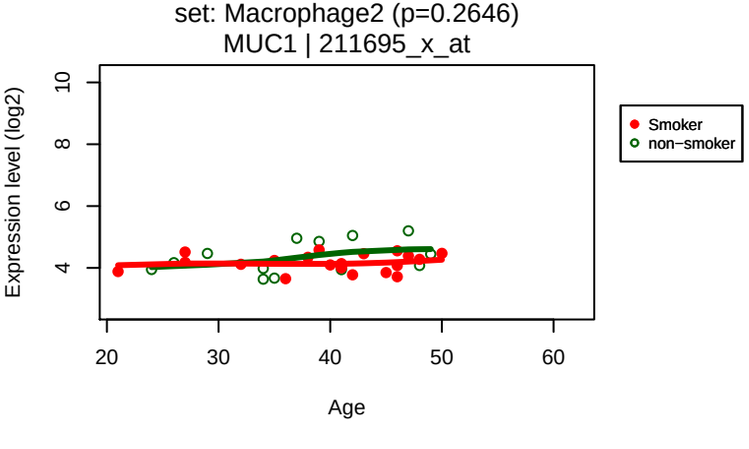

set: SmallAirway2 (p=0.2149)
MUC1 | 211695_x_at
Expression level (log2)
Age
Smoker
non−smoker

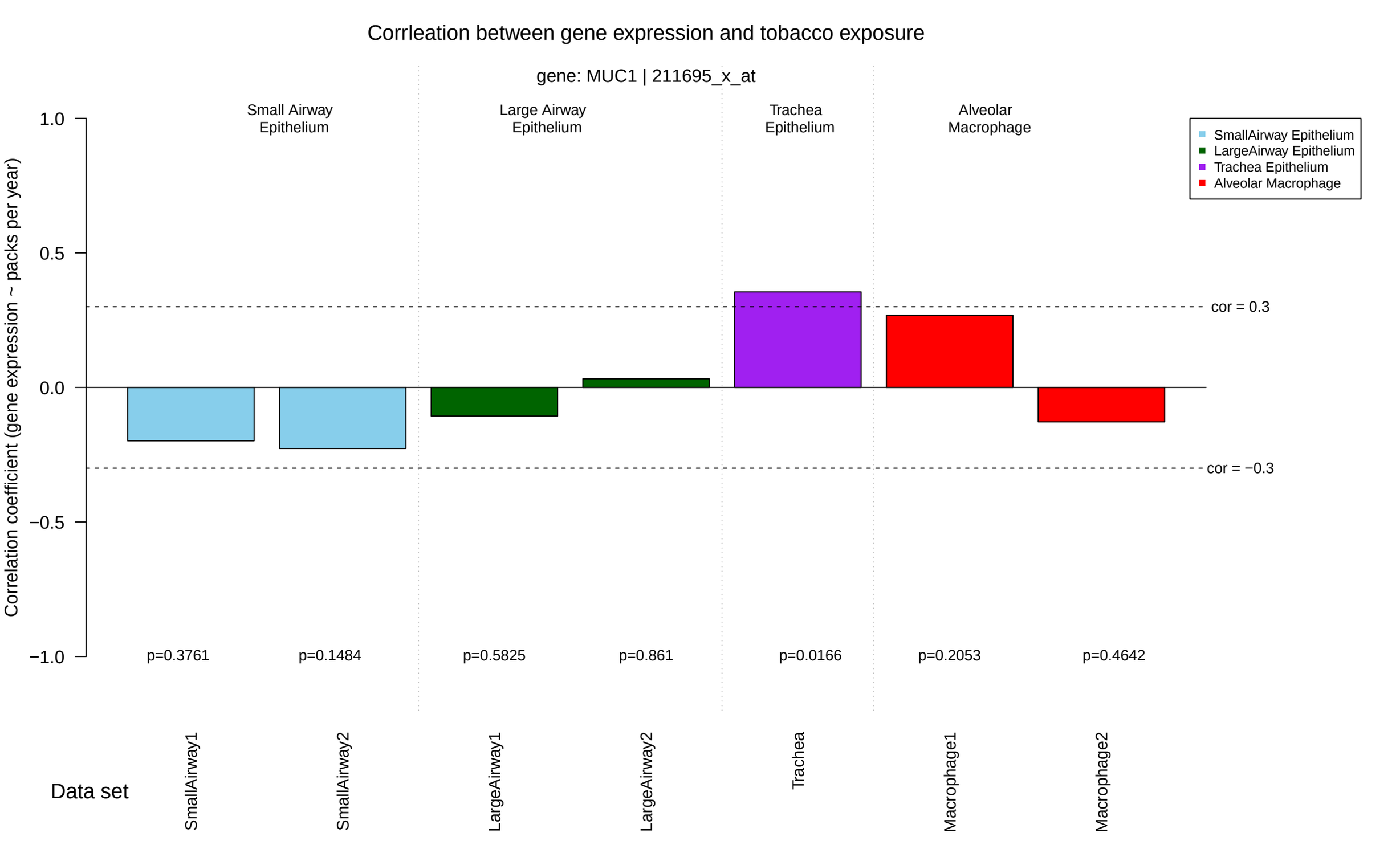

set: LargeAirway1 (p=0.9416)
MUC1 | 211695_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3316)
MUC1 | 211695_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0016)
MUC1 | 211695_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.02)
MUC1 | 211695_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2646)
MUC1 | 211695_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC1 | 211695_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3761
p=0.1484
p=0.5825
p=0.861
p=0.0166
p=0.2053
p=0.4642
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2